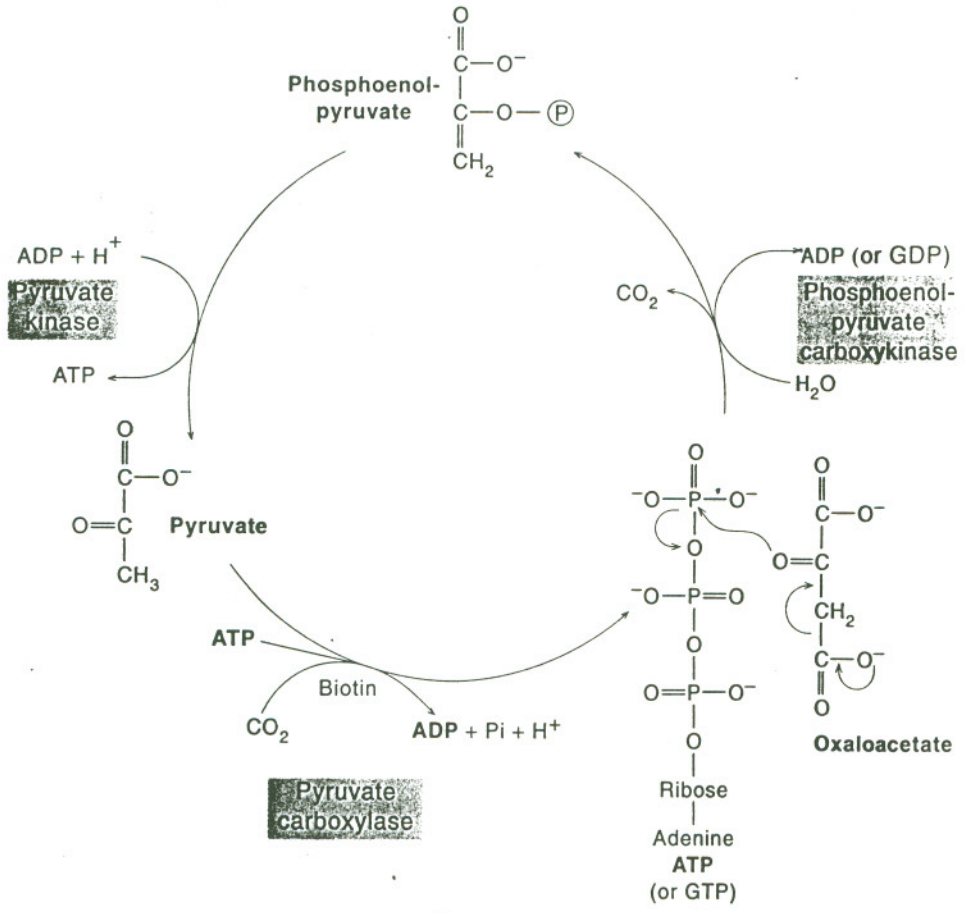
BIOCHEMISTRY CUMULATIVE EXAM

March 1, 1997

A. (40) Write a complete enzymatic pathway for the synthesis of alanine in a liver cell suspension, using glutamate as the <u>sole</u> source of both carbon and nitrogen. You may assume that other metabolites are present in small catalytic amounts.

B. (30) Write a complete balanced equation for the conversion of glutamate to alanine in the liver cell under aerobic conditions, giving the number of CO_2 generated, O_2 consumed, and ATP produced.

C. (30) Determine which carbon atom(s) of alanine will <u>first</u> become labeled if glutamate labeled with C-14 at carbon number 3 is used?



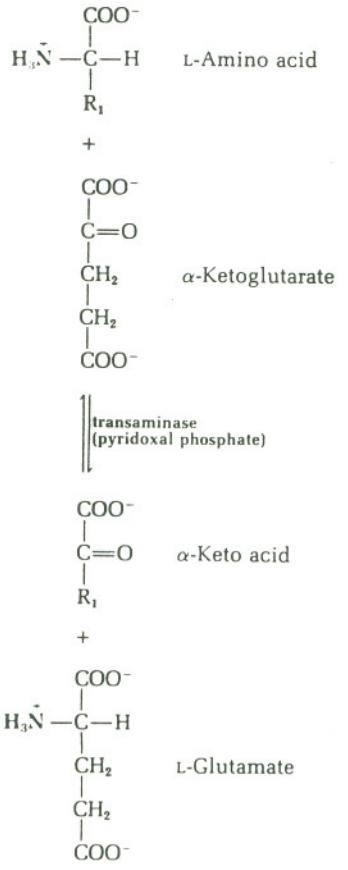
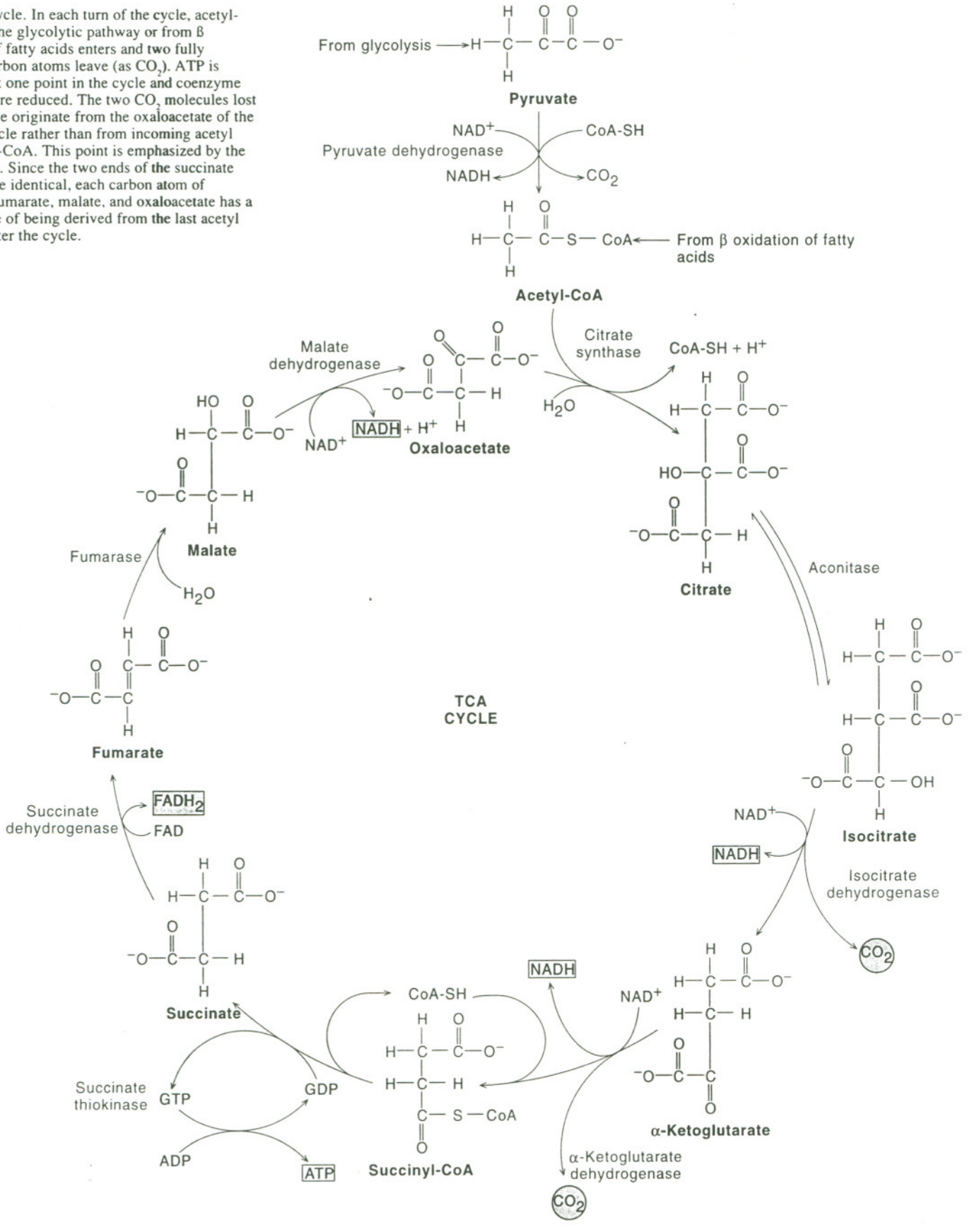


Figure 14.4

The TCA cycle. In each turn of the cycle, acetyl-CoA from the glycolytic pathway or from ß oxidation of fatty acids enters and two fully oxidized carbon atoms leave (as CO_2). ATP is generated at one point in the cycle and coenzyme molecules are reduced. The two CO_2 molecules lost in each cycle originate from the oxaloacetate of the previous cycle rather than from incoming acetyl from acetyl-CoA. This point is emphasized by the use of color. Since the two ends of the succinate molecule are identical, each carbon atom of succinate, fumarate, malate, and oxaloacetate has a 50% chance of being derived from the last acetyl group to enter the cycle.